INCH-POUND

MS16630C
14 March 2012
SUPERSEDING
MS16630B
18 April 1989

DETAIL SPECIFICATION SHEET

RING, RETAINING, EXTERNAL, BEVELED
(TAPERED SECTION TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-R-21248.

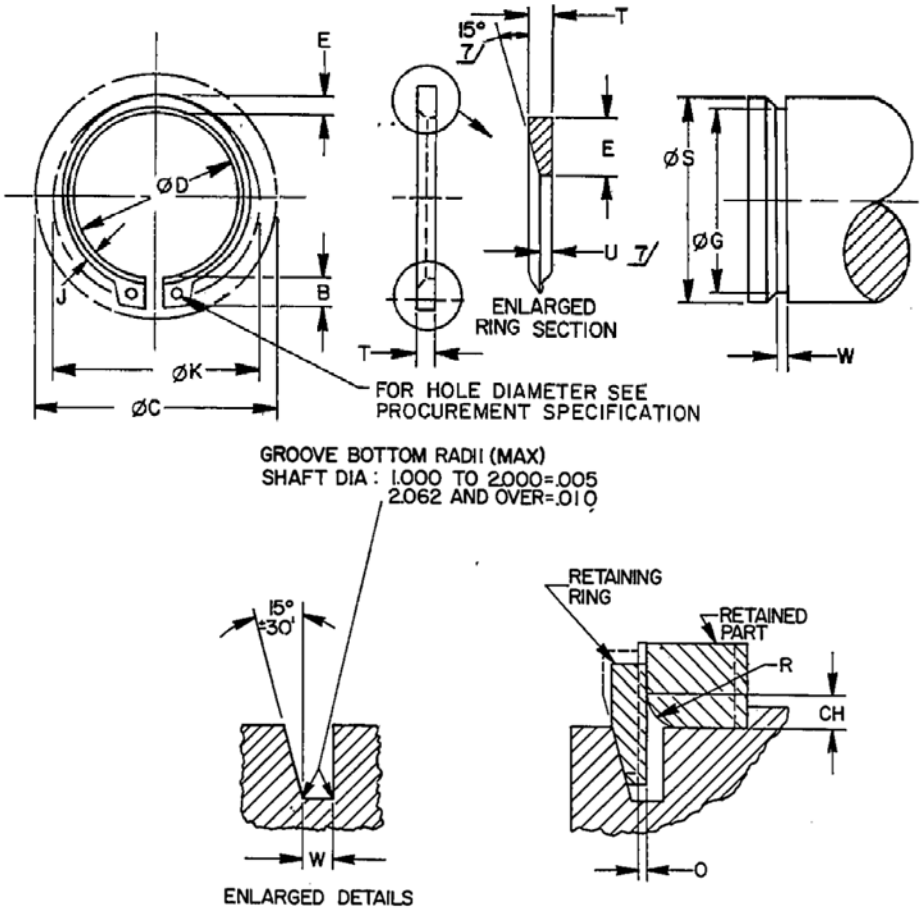


FIGURE 1. RING, RETAINING.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

TABLE I DIMENSIONS.

ØS SHAFT (REF)		ØD FREE		B LUG HEIGHT		E LARGE SECTION HEIGHT		J SMALL SECTION HEIGHT		U 7/ THICKNESS AT BEVELED EDGE		T 1/ THICKNESS		ØG RECOMMENDED GROOVE (REF)		W WIDTH		ØK 3/	ØC 4/ CLEAR	O 5/ TAKE UP	R 6/ OF RETAINED PART (REF)	CH 6/
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MIN		MAX	MAX	MAX
1.000	25.4	.925	+.005	.167		.116	±.005	.065	±.005	.034		.042		.930	+.000	.037		1.144	1.410	.005	.057	.034
1.023	26.0	.946	-.010	.168		.118		,066		.033		.042		.951	-.003	.036		1.170	1.430	.005	.058	.035
															.004 FIM 2/							
1.062	27.0	.982		.181		.122		.069		.041		.050		.992		.044		1.217	1.500	.005	.060	.036
1.125	28.6	1.042		.182		.129		.071		.041		.050		1.051		.044		1.286	1.550	.005	.063	.038
1.188	30.2	1.098		.182		.132		.072		.041		.050		1.108		.044		1.351	1.610	.005	.064	.038
1.250	31.7	1.156		.183		.140		.076		.040		.050		1.166		.043		1.424	1.690	.005	.068	.041
1.312	33.3	1.214	+.010	.183	±.004	.146		.076		.039	±.001	.050	±.002	1.224	+.000	.042	+.002	1.490	1.750	.006	.068	.041
1.375	34.9	1.272	-.015	.184		.152		.082		.039		.050		1.282	-.004	.042	-.000	1.562	1.800	.006	.072	.043
1.438	36.5	1.333		.184		.160		.086		.039		.050		1.343	.005	.042		1.636	1.870	.006	.076	.045
1.500	38.1	1.387		.214		.168		.091		.038		.050		1.397	FIM 2/	.041		1.706	1.990	.007	.079	.047
1.562	39.7	1.446		.235		.172		.093		.049		.062		1.459		.053		1.778	2.100	.007	.082	.049
1.625	41.3	1.503		.235		.180		,097		.049		.062		1.516	+.000	.053		1.849	2.170	.007	.087	.052
1.687	42.9	1.560	+.013	.235		.184	±.006	.099	±.006	.048		.062		1.573	-.005	.052		1.912	2.240	.007	.090	.054
1.750	44.4	1.618	-.020	.237		.188		.101		.048		.062		1.631	.005	.052		1.981	2.310	.008	.091	.054
1.771	45.0	1.637		.237		.190		.102		.048		.062		1.650	FIM 2/	.052		2.004	2.330	.008	.092	.055
1.812	46.0	1.675		.238		.192		.102		.048		.062		1.688		.052		2.047	2.380	.008	.092	.055
1.875	47.6	1.735		.239		.196		.104		.048		.062		1.748		.052		2.114	2.440	.008	.094	.056
1.969	50.0	1.819		.245		.200		.106		.047		.062		1.832		.051		2.209	2.540	.009	.094	.056
2.000	50.8	1.850		.239		.204		.108		.047		.062		1.863		.051		2.246	2.550	.009	.096	.057
2.062	52.4	1.906		.266		.208		.111		.062		.078		1.921		.067		2.315	2.680	.009	.098	.059
2.125	54.0	1.964		.266		.212		.113		.062		.078		1.979		.067		2.386	2.750	.010	.098	.059
2.156	54.8	1.993	+.015	.266		.212	±.007	.113	±.007	.062	±.0015	.078		2.008		.067		2.410	2.780	.010	.097	.058
2.250	57.1	2.081	-.025	.267		.220		.116		.061		.078		2.096		.066		2.513	2.870	.010	.100	.060
2.312	58.7	2.139		.267		.222		.118		.060		.078		2.154		.065		2.577	2.940	.010	.100	.060
2.375	60.3	2.197		.267		.224		.119		.060		.078		2.212		.065		2.640	3.010	.011	.100	.060
2.437	61.9	2.255		.268		.228		.120		.060		.078		2.270		.065		2.706	3.070	.011	.102	.061
2.500	63.5	2.313		.268		.232		.122		.059		.078		2.328		.064		2.772	3.120	.011	.104	.062
2.559	65.0	2.377		.268		.238		.125		.059		.078		2.397		.064		2.845	3.180	.011	.108	.065
2.625	66.7	2.428		.268		.242		.127		.059		.078		2.448		.064		2.910	3.230	.011	.109	.066
2.687	68.3	2.485		.268		.246		.129		.059		.078		2.505		.064		2.975	3.320	.012	.111	.067
2.750	69.8	2.543		.310		.248		.131		.073		.093		2.563		.079		3.041	3.440	.012	.112	.067
2.875	73.0	2.659		.308	±.005	.256		.133		.072		.093		2.679		.078		3.172	3.570	.013	.115	.069
2.937	74.6	2.717	+.020	.308		.260		.136		.072		.093		2.737		.078		3.239	3.640	.013	.116	.070
3.000	76.2	2.775	-.030	.308		.264		.138		.071		.093		2.795	+.000	.077		3.306	3.690	.013	.117	.070
3.062	77.8	2.832		.298		.252		.131		.071	±.002	.093	±.003	2.852	-.006	.077	+.002 5 -.000	3.347	3.740	.014	.107	.064
															.006							
3.125	79.4	2.892		.308		.272		.141		,070		.093		2.912	FIM 2/	.076		3.439	3.820	.014	.120	.072
3.156	80.2	2.920		.308		.274		.143		.070		.093		2.940		.076		3.469	3.830	.014	.120	.072
3.250	82.5	3.006		.308		.280		.145		.070		.093		3.026		.076		3.571	3.940	.015	.123	.074
3.346	85.0	3.092		.308		.286		.147		.069		.093		3.112		.075		3.669	4.040	.015	.126	.076
3.437	87.3	3.179		.308		.292		.148		.069		.093		3.199		.075		3.767	4.140	.016	.129	.077
3.500	88.9	3.237		.328		.285	±.008	.148	±.008	.084	±.0025	.109		3.257		.091		3.821	4.250	.016	.122	.073
3.543	90.0	3.277		.328		.288		.149		.084		.109		3.297		.091	+.003	3.866	4.290	.016	.123	.074
3.625	92.0	3.352		.328		.296		.153		.083		.109		3.372		.090	-.000	3.956	4.370	.017	.127	.076
3.687	93.7	3.410		.330		.302		.156		.083		.109		3.430		.090		4.026	4.430	.017	.129	.078
3.750	95.2	3.468		.332		.310		.160		.082		.109		3.488		.089		4.098	4.500	.017	.133	.080
3.875	98.4	3.584		.330		.318		.163		.082		.109		3.604		.089		4.229	4.600	.018	.137	.082
3.938	100.0	3.642		.342		.318		.163		.081		.109		3.662		.088		4.290	4.700	.018	.137	.082
4.000	101.6	3.700		.352		.318		.163		.081		.109		3.720		.088		4.350	4.780	.019	.135	.081

1/ T - THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e. ±.002 SHOULD BE +.004/-.002.

2/ FIM - (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE SHAFT.

3/ K - MAXIMUM DIAMETER WHEN THE RING IS PROPERLY SEATED AT THE MINIMUM POSITION HALF WAY DOWM IN THE GROOVE.

4/ C - CLEARANCE DIAMETER WHEN THE RING IS SPRUNG OVER THE SHAFT PRIOR TO INSTALLATION INTO THE GROOVE. (DESIGN REFERENCE DIMENSION).

5/ O - END PLAY TAKE.

6/ R AND CH - RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. THRUST LOADS OF RINGS, RETAINING PARTS WITH CORNER RADII OR CHAMFERS ARE TABULATED ON PAGE 6.

7/ 15° AND U - THESE DIMENSIONS ARE FOR GROOVE LOCATION CALCULATIONS ONLY (SEE (f) ON PAGE 7) AND NOT FOR INSPECTION PURPOSES.

.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE III, CLASS 2 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:

   (a) CARBON SPRING STEEL,GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A568/A568M OR ASTM A684/A684M.
   (b) CORROSION RESISTANT STEEL IN ACCORDANCE WITH AMS 5520 (UNS S15700).
   (c) BERYLLIUM COPPER, ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194.

3. HARDNESS

TABLE II. HARDNESS.

ØSHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
1.000 TO 1.023 INCL	47-53 HRC	44-51 HRC	56.5-62 HR30N
1.062 TO 3.437 INCL	47-52 HRC	44-51 HRC	37-43 HRC
3.500 AND OVER	45 TO 50 HRC	44-51 HRC	---

4. PROTECTIVE FINISH OR TREATMENT:

   (a) CARBON STEEL – SHALL BE AS SPECIFIED (SEE TABLE III OR IV):

      (1) CADMIUM PLATE IN ACCORDANCE WITH SAE AMS-QQ-P-416, TYPE II, CLASS 3 OR ASTM B696, TYPE II, CLASS 5.

      (2) ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn5 OR ASTM B695, TYPE II, CLASS 5.

      (3) PHOSPHATE COAT IN ACCORDANCE WITH MIL-DTL-16232, TYPE Z, CLASS II.

   (b) CORROSION RESISTANT STEEL – SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH SAE AMS2700.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III AND IV.

   EXAMPLE: MS16630-1100 IS THE PART NUMBER FOR A CARBON STEEL, CADMIUM PLATED, EXTERNAL, BEVELED RETAINING RING FOR USE IN A 1.000 DIAMETER SHAFT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

TABLE III. DASH NUMBERS FOR MS16630.

ØS SHAFT (REF)	1/ CARBON STEEL CADMIUM PLATE	1/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT	STEEL CORROSION RESISTANT	1/ BERYLLIUM COPPER
	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO
1.000	-1100	-2100	-3100	-4100	-5100
1.023	-1102	-2102	-3102	-4102	-5102
1.062	-1106	-2106	-3106	-4106	-5106
1.125	-1112	-2112	-3112	-4112	-5112
1.188	-1118	-2118	-3118	-4118	-5118
1.250	-1125	-2125	-3125	-4125	-5125
1.312	-1131	-2131	-3131	-4131	-5131
1.375	-1137	-2137	-3137	-4137	-5137
1.438	-1143	-2143	-3143	-4143	-5143
1.500	-1150	-2150	-3150	-4150	-5150
1.562	-1156	-2156	-3156	-4156	-5156
1.625	-1162	-2162	-3162	-4162	-5162
1.687	-1168	-2168	-3168	-4168	-5168
1.750	-1175	-2175	-3175	-4175	-5175
1.771	-1177	-2177	-3177	-4177	-5177
1.812	-1181	-2181	-3181	-4181	-5181
1.875	-1187	-2187	-3187	-4187	-5187
1.969	-1196	-2196	-3196	-4196	-5196
2.000	-1200	-2200	-3200	-4200	-5200
2.062	-1206	-2206	-3206	-4206	
2.125	-1212	-2212	-3212	-4212	
2.156	-1215	-2215	-3215	-4215	
2.250	-1225	-2225	-3225	-4225	
2.312	-1231	-2231	-3231	-4231	
2.375	-1237	-2237	-3237	-4237	

ØS SHAFT (REF)	1/ CARBON STEEL CADMIUM PLATE	1/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT	STEEL CORROSION RESISTANT
	DASH NO.	DASH NO.	DASH NO.	DASH NO.
2.437	-1243	-2243	-3243	-4243
2.500	-1250	-2250	-3250	-4250
2.559	-1255	-2255	-3255	-4255
2.625	-1262	-2262	-3262	-4262
2.687	-1268	-2268	-3268	-4268
2.750	-1275	-2275	-3275	-4275
2.875	-1287	-2287	-3287	-4287
2.937	-1293	-2293	-3293	-4293
3.000	-1300	-2300	-3300	-4300
3.062	-1306	-2306	-3306	-4306
3.125	-1312	-2312	-3312	-4312
3.156	-1315	-2315	-3315	-4315
3.250	-1325	-2325	-3325	-4325
3.346	-1334	-2334	-3334	-4334
3.437	-1343	-2343	-3343	-4343
3.500	-1350	-2350	-3350	-4350
3.543	-1354	-2354	-3354	-4354
3.625	-1362	-2362	-3362	-4362
3.687	-1368	-2368	-3368	-4368
3.750	-1375	-2375	-3375	-4375
3.875	-1387	-2387	-3387	-4387
3.938	-1393	-2393	-3393	-4393
4.000	-1400	-2400	-3400	-4400

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

TABLE IV. SUBSTITUTION TABLE (CROSS REFERENCE OF PART NUMBERS).

ØS SHAFT (REF)	INACTIVE CARBON STEEL	SUBSTITUTE 1/ CARBON STEEL CADMIUM PLATED	SUBSTITUTE 1/ CARBON STEEL ZINC PLATED	SUBSTITUTE 1/ CARBON STEEL PHOSPHATE COATED
	MS16630	MS16630	MS16630	MS16630
1.000	-100	-1100	-2100	-3100
1.023	-102	-1102	-2102	-3102
1.062	-106	-1106	-2106	-3106
1.125	-112	-1112	-2112	-3112
1.188	-118	-1118	-2118	-3118
1.250	-125	-1125	-2125	-3125
1.312	-131	-1131	-2131	-3131
1.375	-137	-1137	-2137	-3137
1.438	-143	-1143	-2143	-3143
1.500	-150	-1150	-2150	-3150
1.562	-156	-1156	-2156	-3156
1.625	-162	-1162	-2162	-3162
1.687	-168	-1168	-2168	-3168
1.750	-175	-1175	-2175	-3175
1.771	-177	-1177	-2177	-3177
1.812	-181	-1181	-2181	-3181
1.875	-187	-1187	-2187	-3187
1.969	-196	-1196	-2196	-3196
2.000	-200	-1200	-2200	-3200
2.062	-206	-1206	-2206	-3206
2.125	-212	-1212	-2212	-3212
2.156	-215	-1215	-2215	-3215
2.250	-225	-1225	-2225	-3225
2.312	-231	-1231	-2231	-3231
2.375	-237	-1237	-2237	-3237
2.437	-243	-1243	-2243	-3243
2.500	-250	-1250	-2250	-3250
2.559	-255	-1255	-2255	-3255
2.625	-262	-1262	-2262	-3262
2.687	-268	-1268	-2268	-3268
2.750	-275	-1275	-2275	-3275
2.875	-287	-1287	-2287	-3287
2.937	-293	-1293	-2293	-3293
3.000	-300	-1300	-2300	-3300
3.062	-306	-1306	-2306	-3306
3.125	-312	-1312	-2312	-3312
3.156	-315	-1315	-2315	-3315
3.250	-325	-1325	-2325	-3325
3.346	-334	-1334	-2334	-3334
3.437	-343	-1343	-2343	-3343
3.500	-350	-1350	-2350	-3350
3.543	-354	-1354	-2354	-3354
3.625	-362	-1362	-2362	-3362
3.687	-368	-1368	-2368	-3368
3.750	-375	-1375	-2375	-3375
3.875	-387	-1387	-2387	-3387
3.938	-393	-1393	-2393	-3393
4.000	-400	-1400	-2400	-3400

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

NOTES:

1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. RECOMMENDED DESIGN LIMITATIONS AND USAGE:

(a) INTENDED USE- TO PROVIDE SHOULDERS FOR POSITIONING AND RETAINING MACHINE COMPONENTS ON SHAFTS. TAPERED DESIGN PRINCIPLE PERMITS RINGS TO MAINTAIN PRACTICTICALLY CONSTANT CIRCULARITY WITHIN THE LIMITS OF EXPANSION IN NORMAL USE. THE RINGS WITH BEVEL ON INNER CIRCUMFERENCE, WHEN SPRUNG INTO GROOVE WITH TAPERED OUTER WALL CORRESPONDING TO RING BEVEL WILL SELF ADJUST AND PROVIDE SECURE PRESSURE FIT AXIALLY TO TAKE UP END-PLAY. THE USE OF THE FOLLOWING FORMULAS ARE BASED ON THE FACT THAT THE RING MATERIAL WILL NOT FAIL IN COMPRESSION.

LIMITATION ON USE – THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING: RINGS SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE.

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS. (ABUTTING COMPONENTS TO HAVE SHARP CORNERS).

$$P = \frac{\pi STX}{F}$$

WHERE:

P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = RING THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY

A FACTOR OF SAFETY, F=4, IS RECOMMENDED, SINCE THE RING UNDER LOAD IS SUBJECTED NOT ONLY TO PURE SHEAR STRESSED BUT ALSO TO BENDING STRESSES.

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL =

$$P = \frac{\pi S \, 2 \, Y_{\frac{d}{2}/}}{F}$$

WHERE:

P = ALLOWABLE COMPRESSION LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
d = BASIC GROOVE DEPTH OF BEVELED GROOVE (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY

TO INSURE A SAFE WORKING LOAD, A SAFETY FACTOR, F = 2, IS RECOMMENDED.

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF HOUSING =

$$Z = 1.5d$$

WHERE:

Z = MINIMUM DISTANCE BEWTWEEN OUTER GROOVE WALL AND END OF SHAFT (INCHES)
d = GROOVE DEPTH (INCHES)

(e) DEFLECTION, (UNDER LOAD) =

$$\Delta_L = \frac{L}{E\frac{d}{2}}$$

d = GROOVE DEPTH (INCHES)
E = YOUNG'S MODULUS OF ELASTICITY OF GROOVE MATERIAL
L = APPLIED LOAD (POUNDS)
Δ_L = DEFLECTION, AT LOAD "L" (INCHES)

1/ X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS.OF CARBON STEEL AND CORROSION RESISTANT STEEL.
X = 100,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF BERYLLIUM COPPER.
2/ $\frac{d}{2}$ = HALF OF THE GROOVE DEPTH OF THE BEVELED GROOVE WILL MAINTAIN THE MINIMUM CONTACT AREA OF RING IN GROOVE. ALLOWABLE LOAD CALCULATION SHOULD BE BASED ON $\frac{d}{2}$.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

(f) LOCATION OF GROOVE =

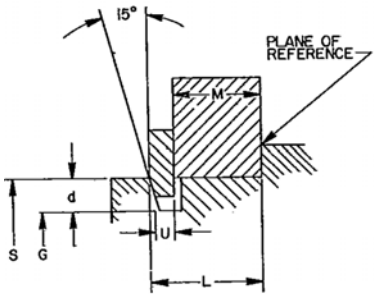
(1) MINIMUM INSERTION OF RING IN GROOVE -

$L_{MIN} \geq {}^{M}_{MAX} + {}^{U}_{MAX}$ + TAKE-UP (AS LISTED)

(2) MAXIMUM INSERTION OF RING IN GROOVE –

$L_{MAX} \leq {}^{M}_{MIN} + {}^{U}_{MIM}$ + 2 TIMES TAKE-UP (AS LISTED)

(3) TAKE-UP (IN ORDER TO FUNCTION PROPERLY THE RING TAKE-UP SHOULD EXCEED THE SUM TOTAL OF THE TOLERANCES.)



TAKE-UP $> {}^{\Delta}_{L} + {}^{\Delta}_{M} + {}^{\Delta}_{U}$ WHERE: ${}^{\Delta}_{L} = {}^{L}_{MAX} - {}^{L}_{MIN}$

${}^{\Delta}_{M} = {}^{M}_{MAX} - {}^{M}_{MIN}$

${}^{\Delta}_{U} = {}^{U}_{MAX} - {}^{U}_{MIN}$

(g) VIBRATION = A RETAINING RING WILL BE SECURE AGAINST VIBRATION IF ITS SHEAR FATIGUE STRENGTH EQUALS OR EXCEEDS THE FORCE CREATED BY THE MASS OF THE PART BEING REPEATEDLY ACCELERATED AGAINST THE RING.

(1) SHEAR STRENGTH OF RING (CRITICAL). (ABUTTING COMPONENTS TO HAVE SHARP EDGES).

$$XT\ 1.1\ S \geq \frac{W}{g} a$$

OR FOR HARMONIC MOTION

$$a = \delta \omega^2$$

$$XT\ 1.1\ S \geq \frac{W}{g} \delta \omega^2$$

WHERE:

X = ULTIMATE SHEAR STRENGTH OF RING MATERIAL (PSI) 3/
T = RING THICKNESS (INCHES)
S = SHAFT DIAMETER (INCHES)
W = WEIGHT OF ABUTTING PARTS (POUNDS)
g = ACCELERATION DUE TO GRAVITY (IN/SEC^2)
δ = AMPLITUDE OF VIBRATION (INCHES)
a = ACCELERATION (IN.SEC^2)
ω = ANGULAR SPEED (RAD/SEC)

(2) COMPRESSIVE STRENGTH OF GROOVE MATERIAL (CRITICAL) =

$$Y \frac{d}{2} 1.65\ S \geq \frac{W}{g} \delta \omega^2$$

WHERE:

Y = YEALD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL. (PSI)
d = GROOVE DEPTH (INCHES)
OTHER SYMBOLS AS SHOWN IN (1) ABOVE

(h) IMPACT CAPACITY OF RING OR GROOVE WALL =

$I_R = \frac{PT}{2}$ FOR THE RING (INCH POUNDS), ABUTTING COMPONENTS TO HAVE SHARP CORNERS

$I_G = \frac{Pd}{2}$ FOR THE GROOVE (INCH POUNDS).

WHERE:

P = ALLOWABLE THRUST LOAD OF RING OR GROOVE (POUNDS).
T = RING THICKNESS (INCHES).
I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS).
d = GROOVE DEPTH (INCHES).
I_R = IMPACT CAPACITY OF RING (INCH POUNDS).

3/ X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .OF STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF BERYLLIUM COPPER.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

(i) LOAD CAPACITY (WITH THE RETAINED PART RADIUSED OR CHAMFERED) =

WHEN THE RADIUS OR CHAMFER OF THE RETAINED PART DOES NOT EXCEED THE MAXIMUM RADIUS ALLOWED FOR THE BOTTOM OF THE RING GROOVE, THE LESSER LOAD CAPACITY COMPUTED FROM THE FORMULAS ON SHEETS 6 AND 8 APPLY. THE CORNER RADII AND CHAMFERS LISTED ON SHEETS 2 WERE CHOSEN AS LARGE AS POSSIBLE FOR THE RING SIZES INVOLVED AND ARE RELATED TO THE MAXIMUM THRUST LOADS LISTED IN TABLE V.

IF THE CORNER RADII OR CHAMFERS ARE SMALLER THAN THOSE LISTED, THEN THE THRUST LOADS INCREASE PROPORTIONALLY, IN ACCORDANCE WITH THE FOLLOWING FORMULAS:

$P^1 = \frac{P\,CH}{CH^1}$ OR $P^1 = \frac{PR}{R^1}$

WHERE:

P^1 = NEW ALLOWABLE THRUST LOAD.
P = LISTED ALLOWABLE THRUST LOAD.
CH^1 = NEW (SMALLER) CHAMFER.
CH = LISTED CHAMFER.
R^1 = NEW (SMALLER) CORNER RADIUS.
R = LISTED CORNER RADIUS.

LIMIT LOADS LISTED IN TABLE V ARE BASED ON RINGS OF CARBON STEEL OR CORROSION RESISTANT STEEL (WORKING STRESS = 250,000 P.S.I.) AND OF BERYLLIUM COPPER (WORKING STRESS = 180,000 P.S.I.). IF THE ALLOWABLE GROOVE CAPACITY LOADS, AS CALCULATED BY USING THE FORMULA ON PAGES 6, 7 AND 8 ARE LESS THAN THEY SHOULD BE USED.

TABLE V. LIMIT LOADS.

NOMINAL RING SIZE		ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITH PARTS HAVING MAXIMUM CORNER RADII OR CHAMFERS.	
FROM	TO	CARBON STEEL OR CORROSION RESISTANT STEEL	BERYLLIUM COPPER
1.000	1.023	1340 LB	970 LB
1.062	1.500	1950 LB	1400 LB
1.562	2.000	3000 LB	2100 LB
2.062	2.687	5000 LB	
2.750	3.437	7350 LB	
3.500	4.000	10500 LB	

4. CHANGES FROM PREVIOUS ISSUE. MARGINAL NOTATIONS ARE NOT USED IN THIS REVISION TO IDENTIFY CHANGES WITH RESPECT TO THE PREVIOUS ISSUE DUE TO THE EXTENT OF THE CHANGES.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST

Custodians:
Army - AR
Navy - OS
Air Force - 99

Preparing activity:
DLA - IS

(Project 5325-2012-003)

Review activities:
Army – AT, AV, CR4, GL, MI
Navy - MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.